CHICAGO BUILDING PERMITS

Investment Opportunity Assessment

Real Estate Developers & Investment Firms

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Today's Analysis

01 The Data

Chicago building permits (2024–present) + neighborhood socioeconomics

02 Where Capital Flows

Geographic concentration across 77 community areas

03 Project Mix

Renovation-heavy market: 82% renovations vs 18% new construction

04 Demographics & Returns

How income levels drive investment activity

05 Predictive Models

Linear & logistic regression to predict project cost and size

06 Recommendations

3-tier investment strategy by capital level and risk tolerance

The Data

PRIMARY DATASET

Chicago Building Permits

Source:	City of Chicago Socrata API
Date Range:	January 2024 – Present
Records:	~60,000+ permits
Key Fields:	Cost, permit type, community area, date

SECONDARY DATASET

Chicago Socioeconomic Indicators

Source:	City of Chicago Socrata API
Coverage:	All 77 community areas
Key Fields:	Per capita income, poverty rate, unemployment
Purpose:	Neighborhood investment context

Where Capital Flows

$3.9B

Near North Side
(#1 by investment)

r = 0.66

Income ↔ Investment
correlation

Top 5

Areas = >50% of
all investment

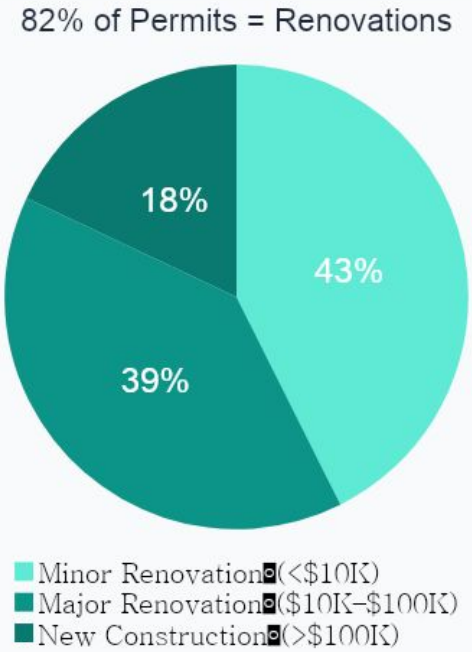
Key Insight

"Money follows money"

High-income neighborhoods dominate investment. North Side & Downtown capture the vast majority of development capital.

Gentrification signal: 8–12 lower-income areas show above-median construction activity.

Project Mix & Capital Requirements



Strategy	Median Cost	% of $ Volume
New Construction	$300K	95.6%
Major Renovation	$24K	4.0%
Minor Renovation	$3K	0.5%

Capital-Intensive

>$500K

New construction in Tier 1 markets (North Side, Loop)

Value-Add

$100K–$500K

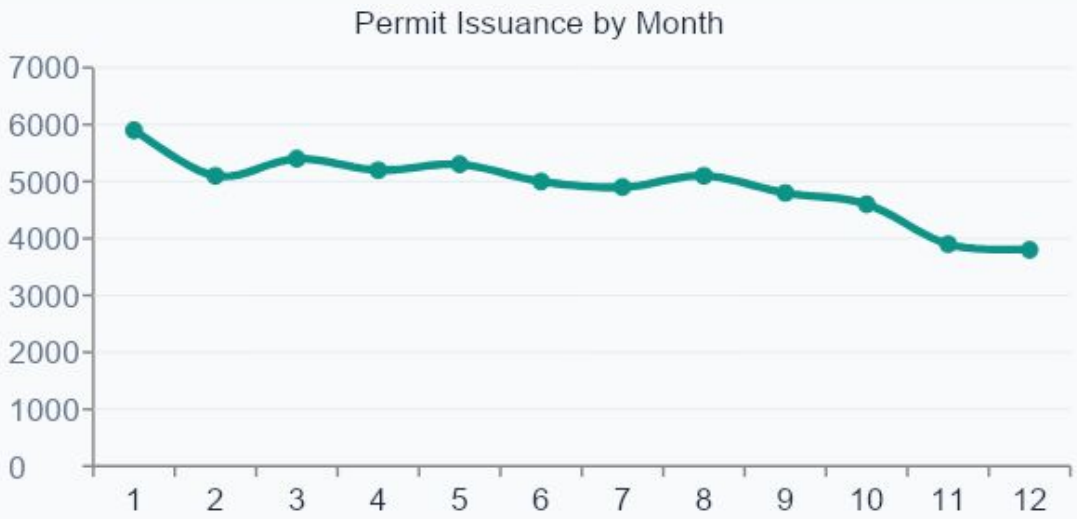
Major renovations in Tier 1 & 2 markets

Volume-Based

<$100K

High-volume minor renos on South/West Side

EDA Highlights



Seasonal Insights

Peak: Jan–May
Contractors plan spring/summer work

Trough: Nov–Dec
35–40% fewer permits issued

Action: Submit permits Nov–Feb for faster processing

Q4 Income Areas

60%+ of all investment

Downtown/Loop

Dwarfs all other regions

Fee Ratio

Consistent across income tiers

Predictive Models

MODEL 1 | LINEAR REGRESSION

Predict log(project cost)

19 features: demographics + permit type + fees + geography + season

R^2	0.387	38.7% of variance in log-cost explained
RMSE	2.029	log-dollar avg error per prediction
MAE	1.416	log-dollar avg absolute error

Top drivers: Renovation type (0.99), New Construction (0.57), total_fee (0.28)

Permit type dummies are the dominant signal. Adding fees + geography lifts R^2 from 0.008 to 0.387 vs demographics-only.

MODEL 2 | LOGISTIC REGRESSION

Classify major projects (reported cost ≥ $100K)

18.2% major class — class_weight='balanced' corrects for imbalance

AUC	0.905	Primary metric — ranks major above minor 90.5% of the time
Acc.	82.6%	Overall accuracy on held-out test set
Prec.	51.2%	1 in 2 predicted-major permits are correct
Recall	78.9%	Catches nearly 4 in 5 actual major projects
F1	62.1%	Strong balanced precision-recall score

ROC-AUC of 0.905 means the model correctly ranks a major project above a minor one 90.5% of the time

Investment Recommendations

TIER 1

Premium Markets

Score > 0.45

Areas	Near North Side, Lincoln Park, Loop, Lake View
Strategy	Major construction & new development
Capital	> $500K per project
Risk	Low

TIER 2

Emerging Opportunities

Score 0.40–0.45

Areas	Near South Side, Edison Park, Beverly
Strategy	Value-add renovations
Capital	$100K – $500K
Risk	Medium

TIER 3

Volume Markets

Score < 0.40

Areas	South Side & West Side
Strategy	High-volume minor renovations
Capital	< $100K per project
Risk	Higher / Operational

Conclusions & Next Steps

Key Takeaways

- Money follows money — r=0.66 income↔investment correlation
- 82% of permits are renovations; new construction drives 95.6% of dollars
- 8–12 gentrification-stage areas offer the best risk-adjusted upside
- Seasonality is real — submit permits Nov–Feb for faster approvals
- Investment score framework enables systematic, data-driven screening

Future Work

- 5–10 year historical permit trends
- Crime & school quality data integration
- Transit proximity analysis
- ML neighborhood forecasting models
- Validate investment score weights with actual ROI data

Thank you · Questions?